

Lusson

**Unobserved
confounders**

**Empirical
researchers**



The Fork



Open unless you
condition on Z

The Pipe



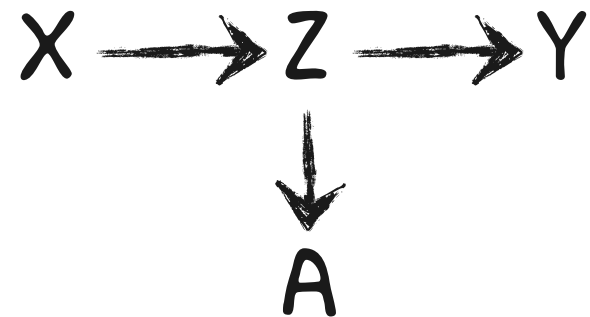
Open unless you
condition on Z

The Collider

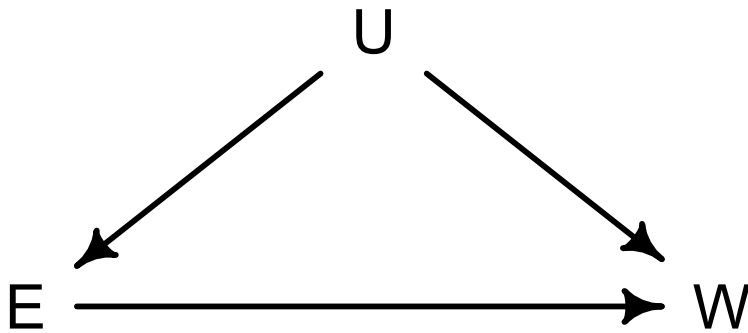


Closed until you
condition on Z

The Descendant



Conditioning on A is
like conditioning on Z



Two paths from E to W:

(1) $E \rightarrow W$

(2) $E \leftarrow U \rightarrow W$

Close 2nd path by conditioning on U, closing the pipe.



Ulysses' Compass

- Two major hazards: (1) Too simple (2) Too complex



Goals

- Understand *overfitting* and *underfitting*
- Introduce *regularization*
- Cross-validation & information criteria:
 - estimate predictive accuracy
 - estimate overfitting risk
 - understand how overfitting relates to complexity
 - identify influential observations
- See that prediction and causal inference are different objectives



The Problem with Parameters

- **What should a model learn from a sample?**
- *Underfitting*: Learning too little from the data. Too simple models both fit and predict poorly.
- *Overfitting*: Learning too much from the data. Complex models tend to fit better, predict worse.
- Want to find a model that navigates between underfitting and overfitting
- Problem: Fit to sample always* improves as we add parameters

*Not true of multilevel models & other types

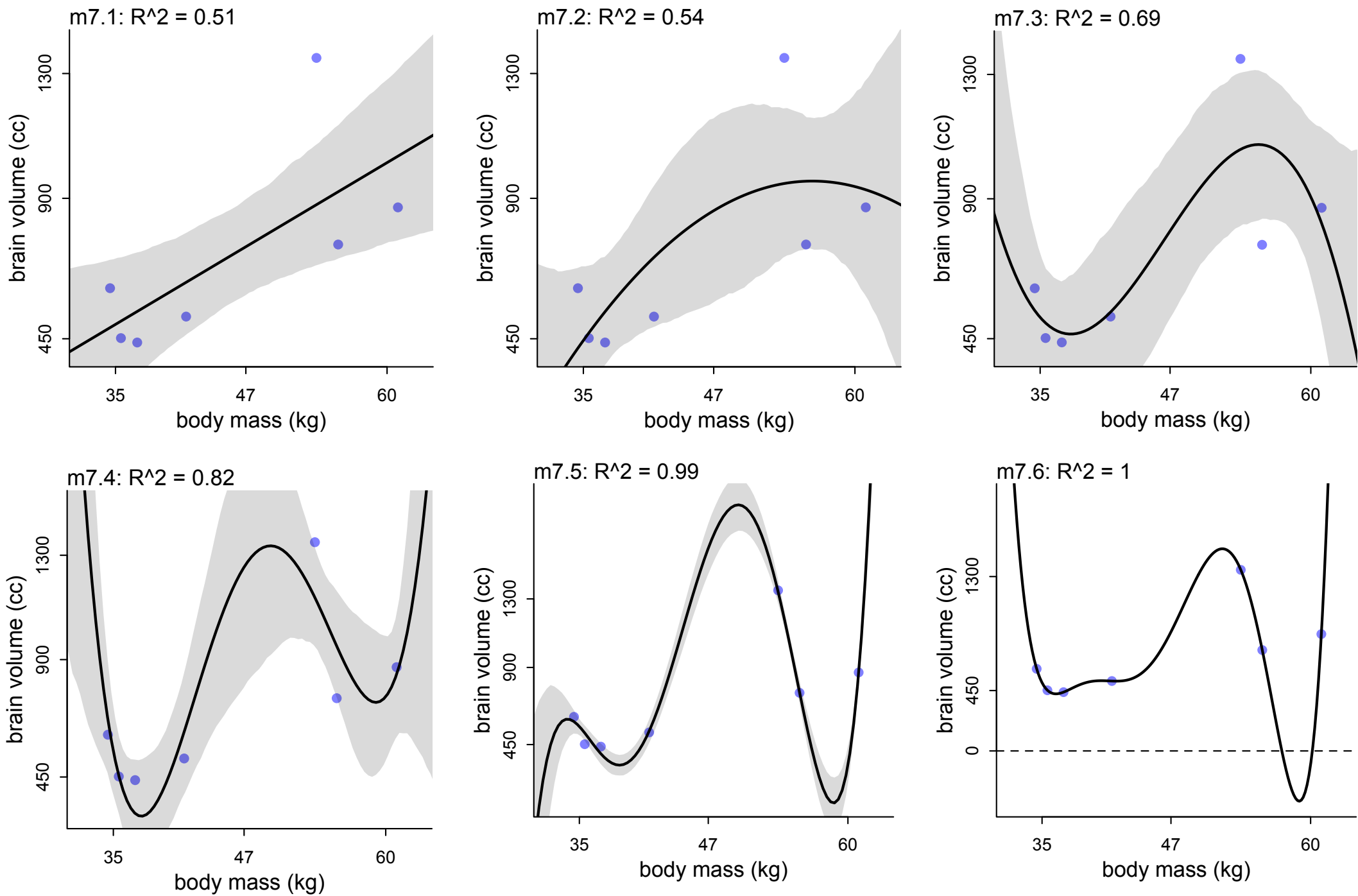
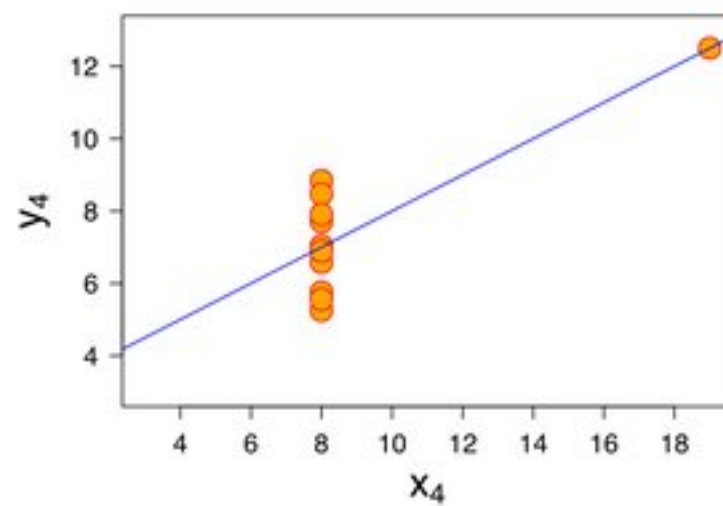
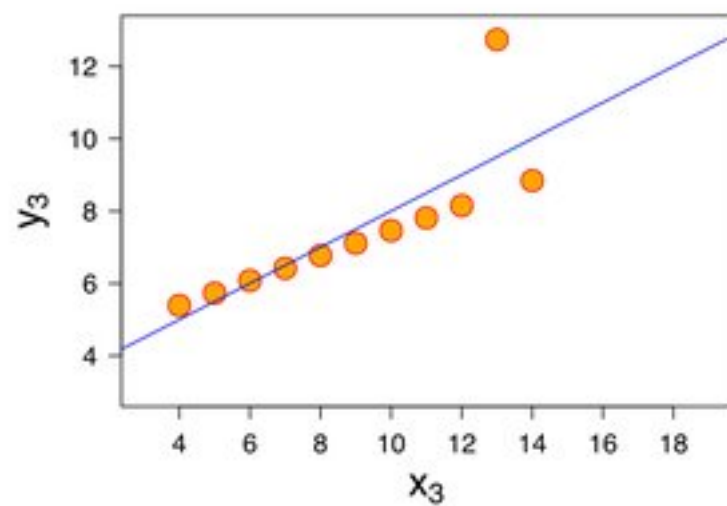
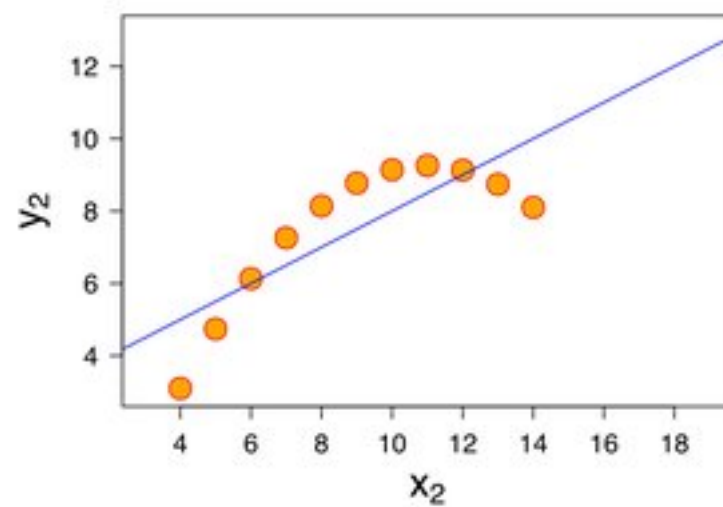
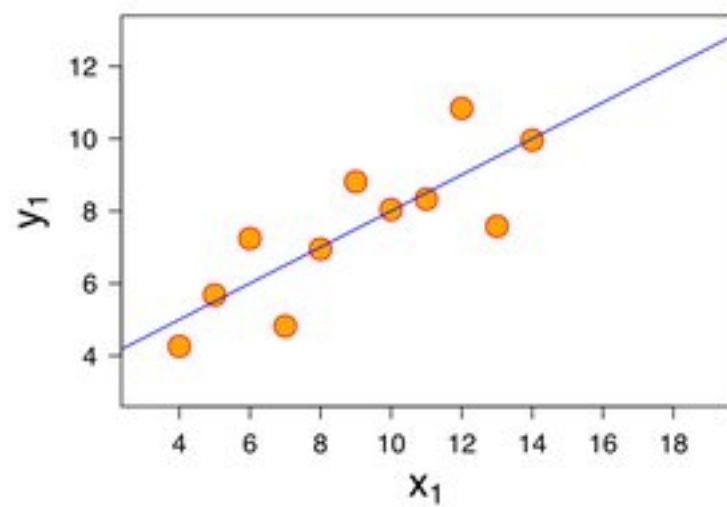
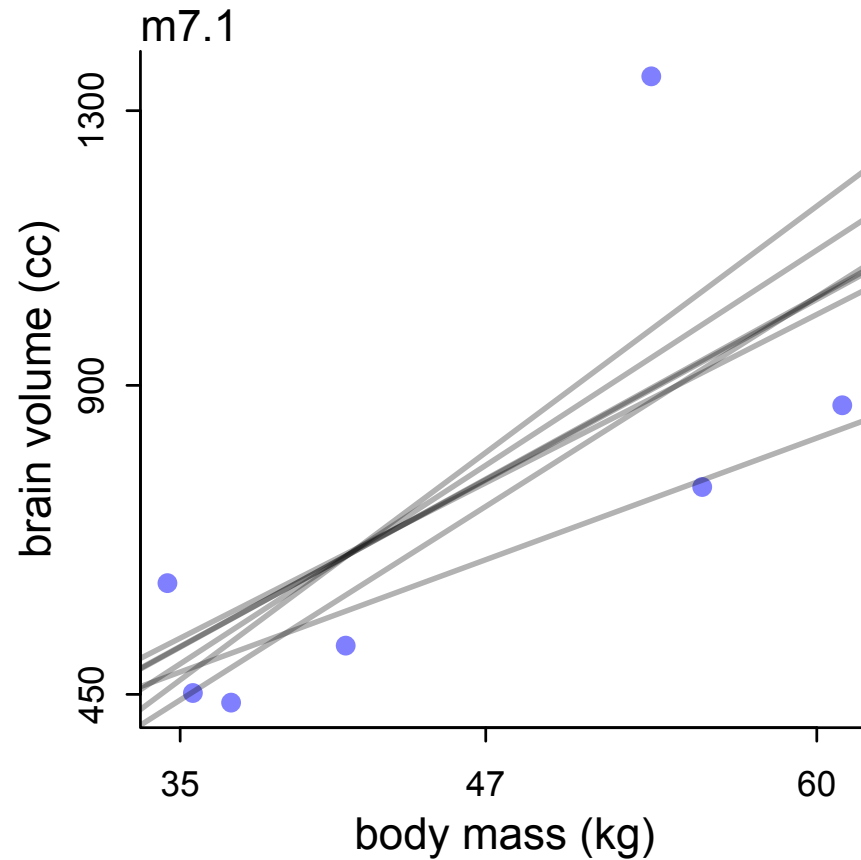


Figure 7.3



Underfitting
Insensitive to
exact data



Overfitting
Very sensitive to
exact data

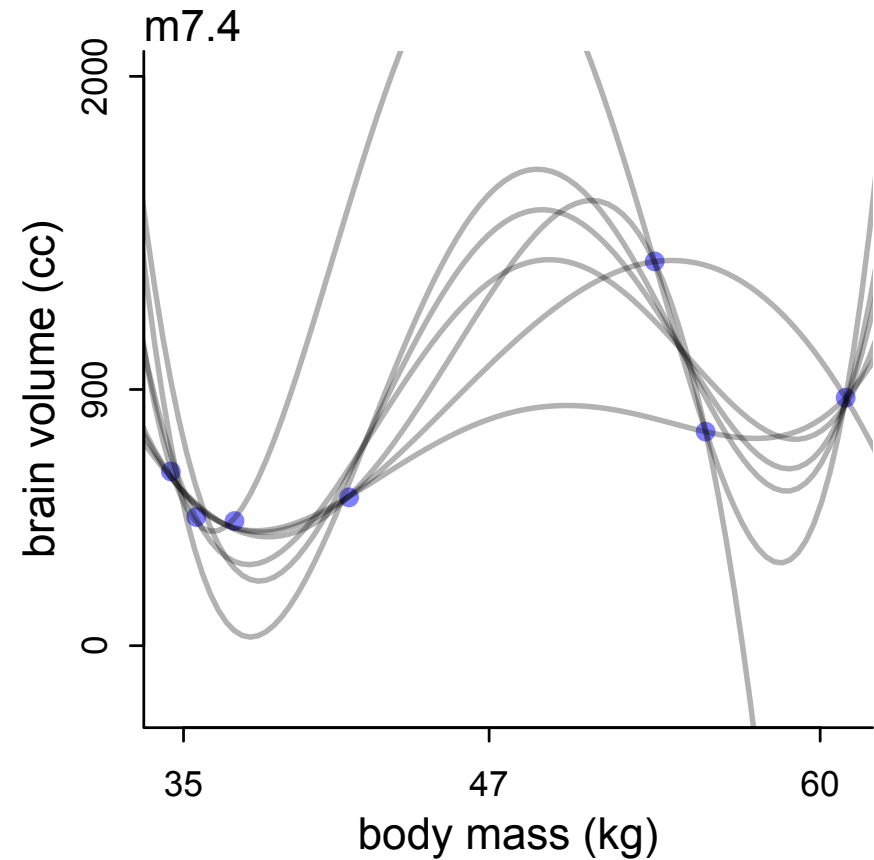
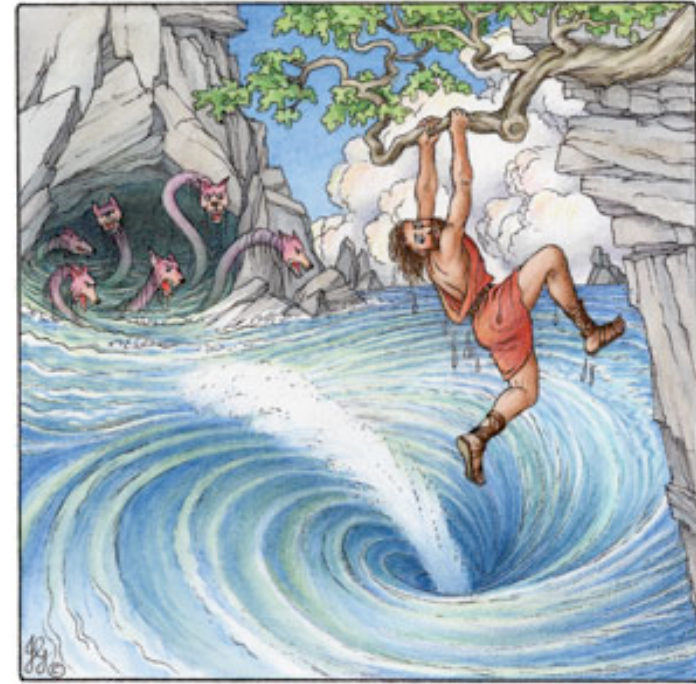


Figure 7.5

Importance of being *regular*

- Want the *regular* features of the sample
- Strategies
 - Regularizing priors (penalized likelihood)
 - Cross-validation
 - Information criteria
 - Science!
- Proper approach depends upon purpose
- Answers are never *only* in the data, but they do usually require data

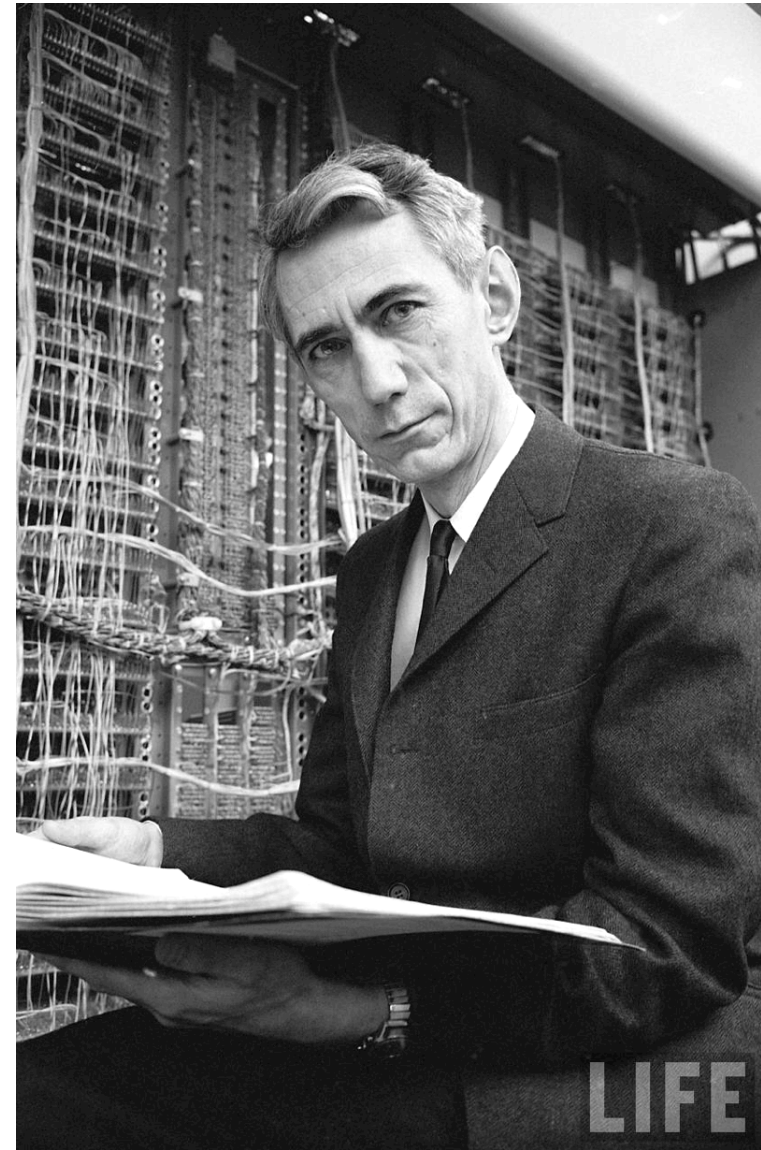


Information entropy

- 1948, Claude Shannon derived *information entropy*:

$$H(p) = -E \log(p_i) = - \sum_{i=1}^n p_i \log(p_i)$$

Uncertainty in a probability distribution is average (minus) log-probability of an event.



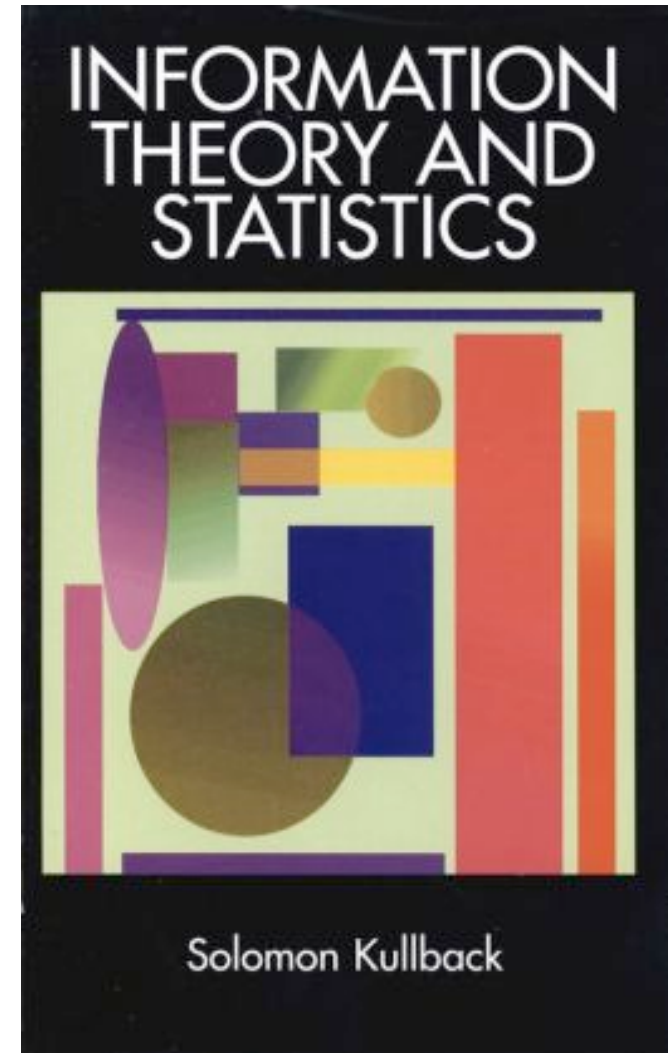
Shannon (1916–2001)

Entropy to accuracy

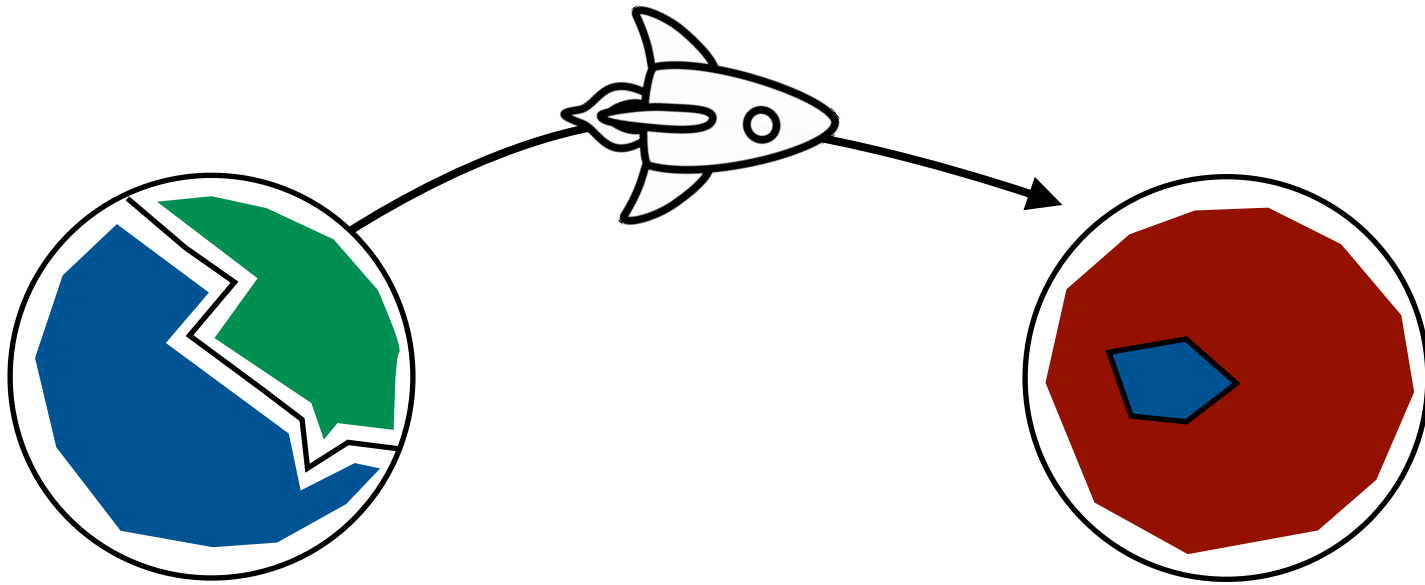
- Two probability distributions: p , q
- p is true, q is model
- How accurate is q , for describing p ?
- Distance from q to p : *Divergence*

$$D_{\text{KL}}(p, q) = \sum_i p_i (\log(p_i) - \log(q_i))$$

Distance from q to p is the average difference in log-probability.



Divergence is not symmetric!



Everybody overfits

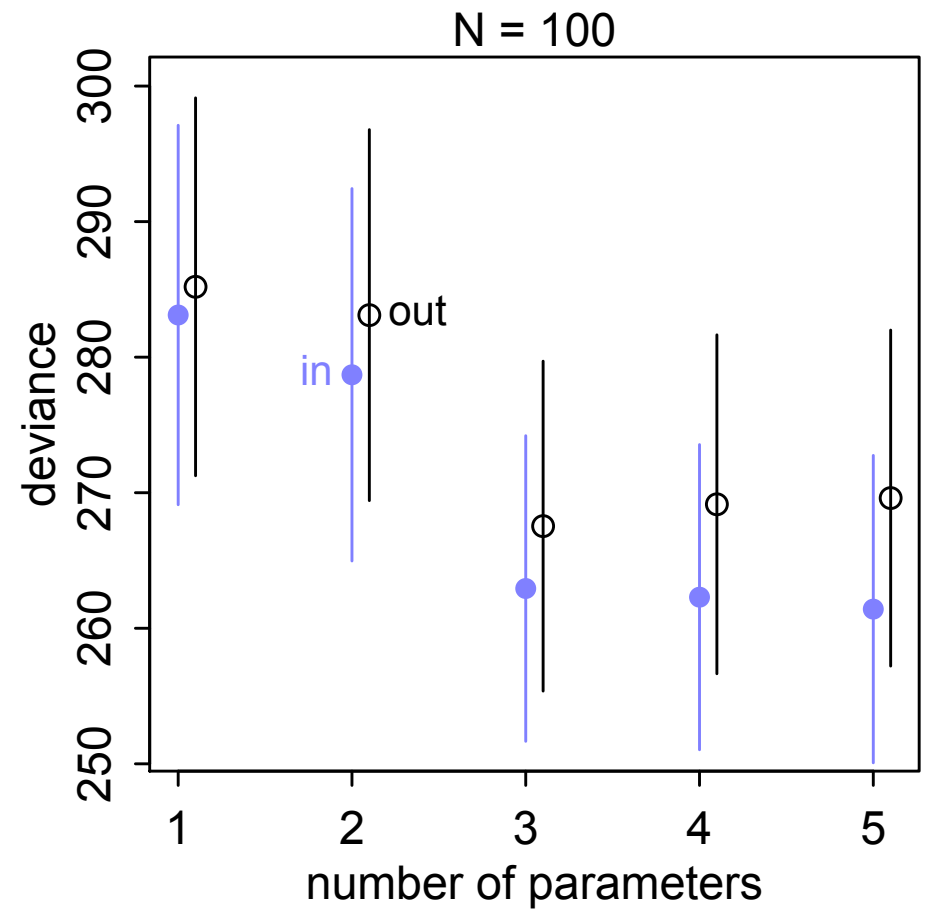
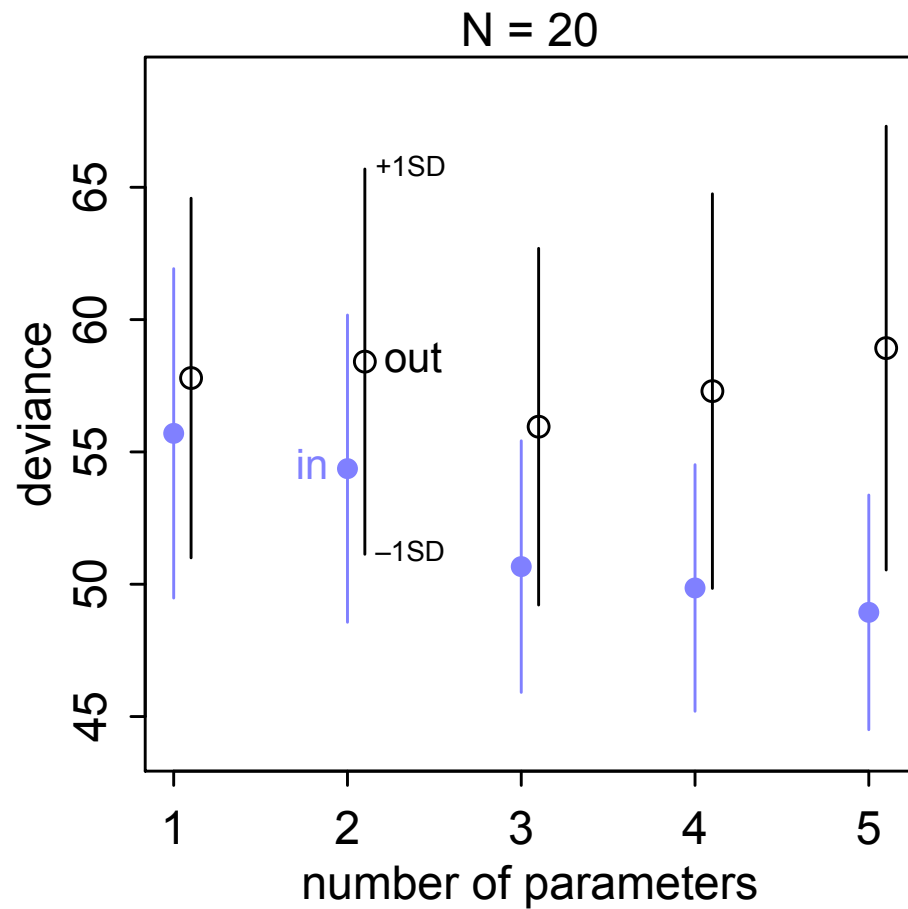


Figure 7.7

Regularization

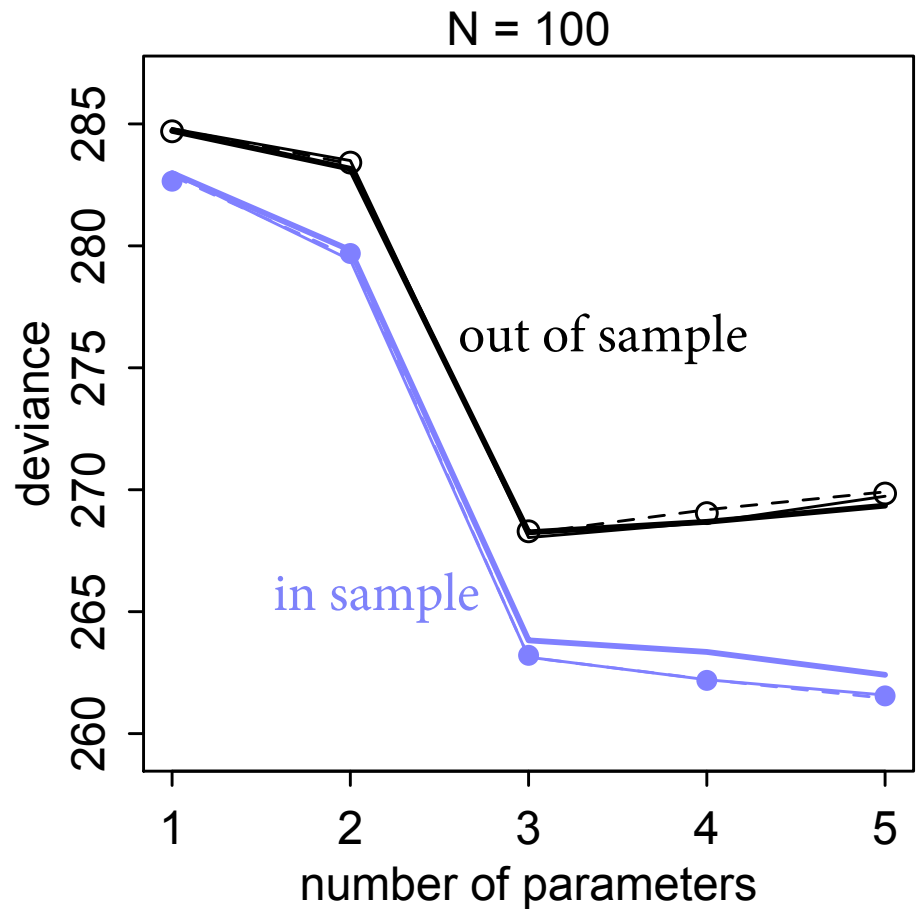
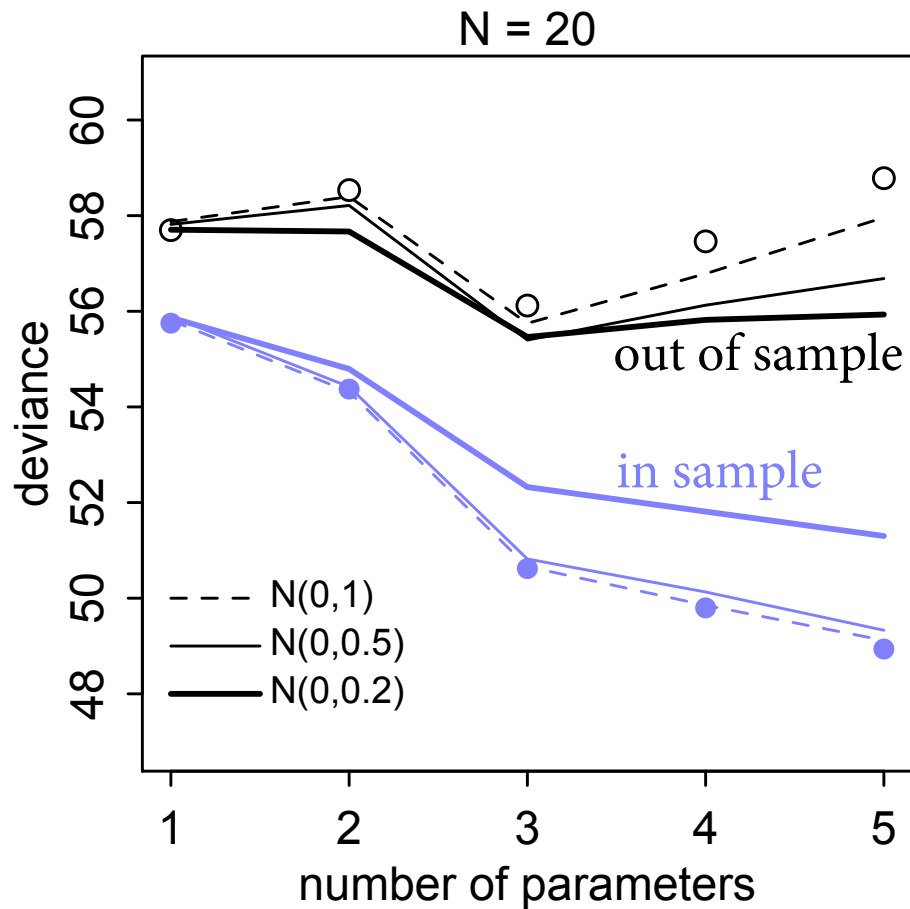


Figure 7.9

Regularization

- **Must be skeptical of the sample!**
- Use informative, conservative priors to reduce overfitting => model learns less from sample
- But if too skeptical, model learns too little
- Such priors are *regularizing*



Smooth Cross-validation

- Most common: Leave-one-out
- Very expensive!
- Useful approximation: Importance sampling (IS)
- More useful: Pareto-smoothed importance sampling (PSIS)
- PSIS-LOO accurate, lots of useful diagnostics
- LOO function in rethinking
- See also `loo` package



Prof Aki Vehtari (Helsinki),
smooth estimator

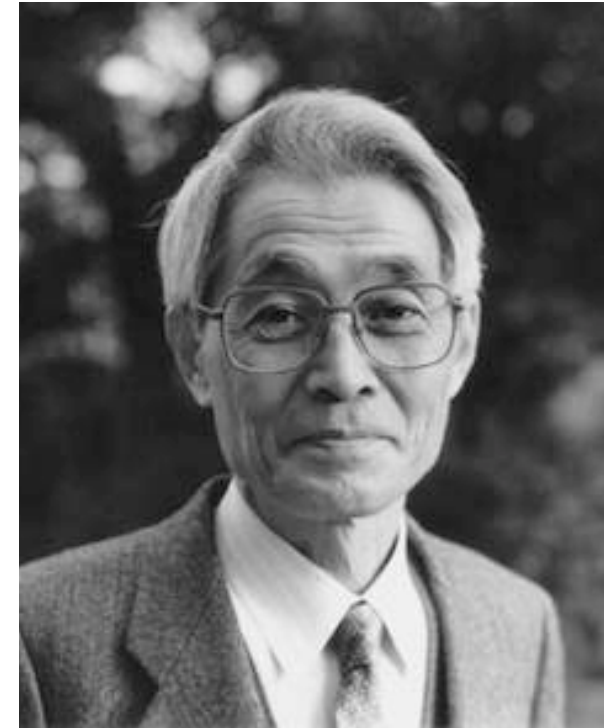
Akaike information criterion

[ah-ka-ee-kay]

- Estimate K-L Distance in theory
- Most famous is the first, AIC
- Under some strict conditions:

$$\text{AIC} = D_{\text{train}} + 2k \approx E D_{\text{test}}$$

k is parameter count



Hirotugu Akaike

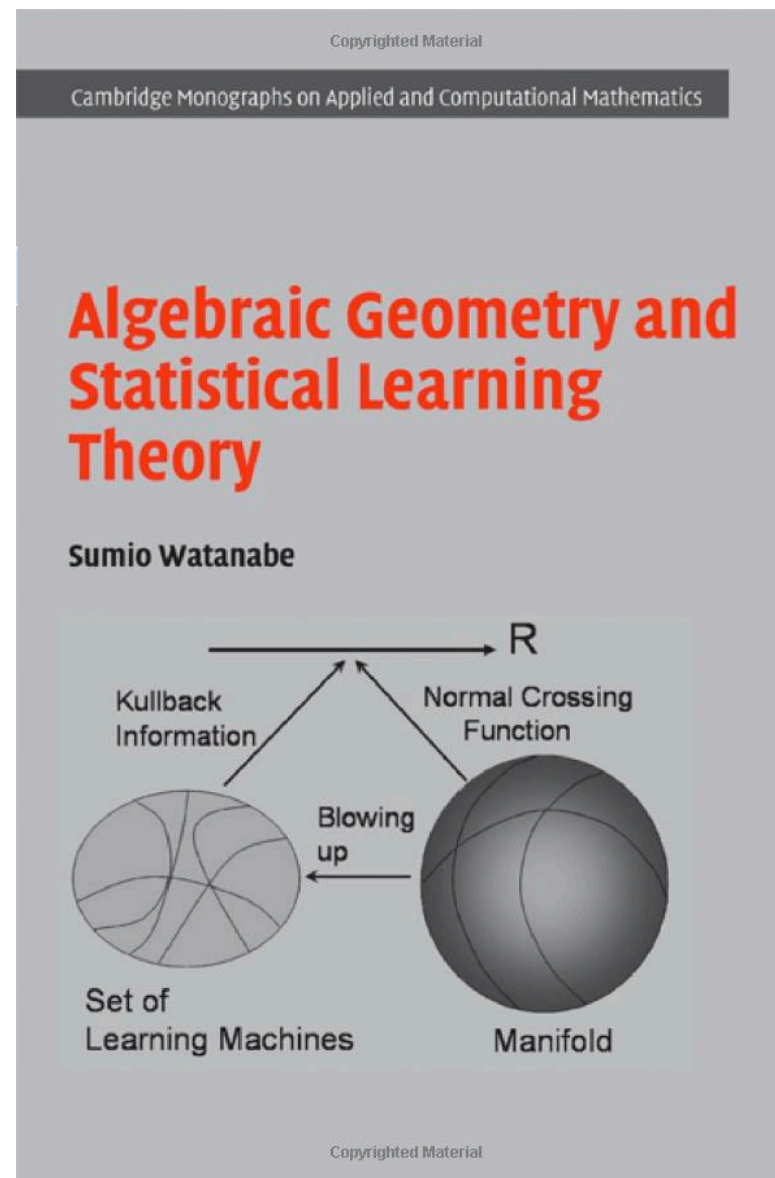
赤池弘次
(1927–2009)

Widely Applicable IC

- AIC of historical interest now
- Widely Applicable Information Criterion (WAIC)
 - Sumio Watanabe 2010

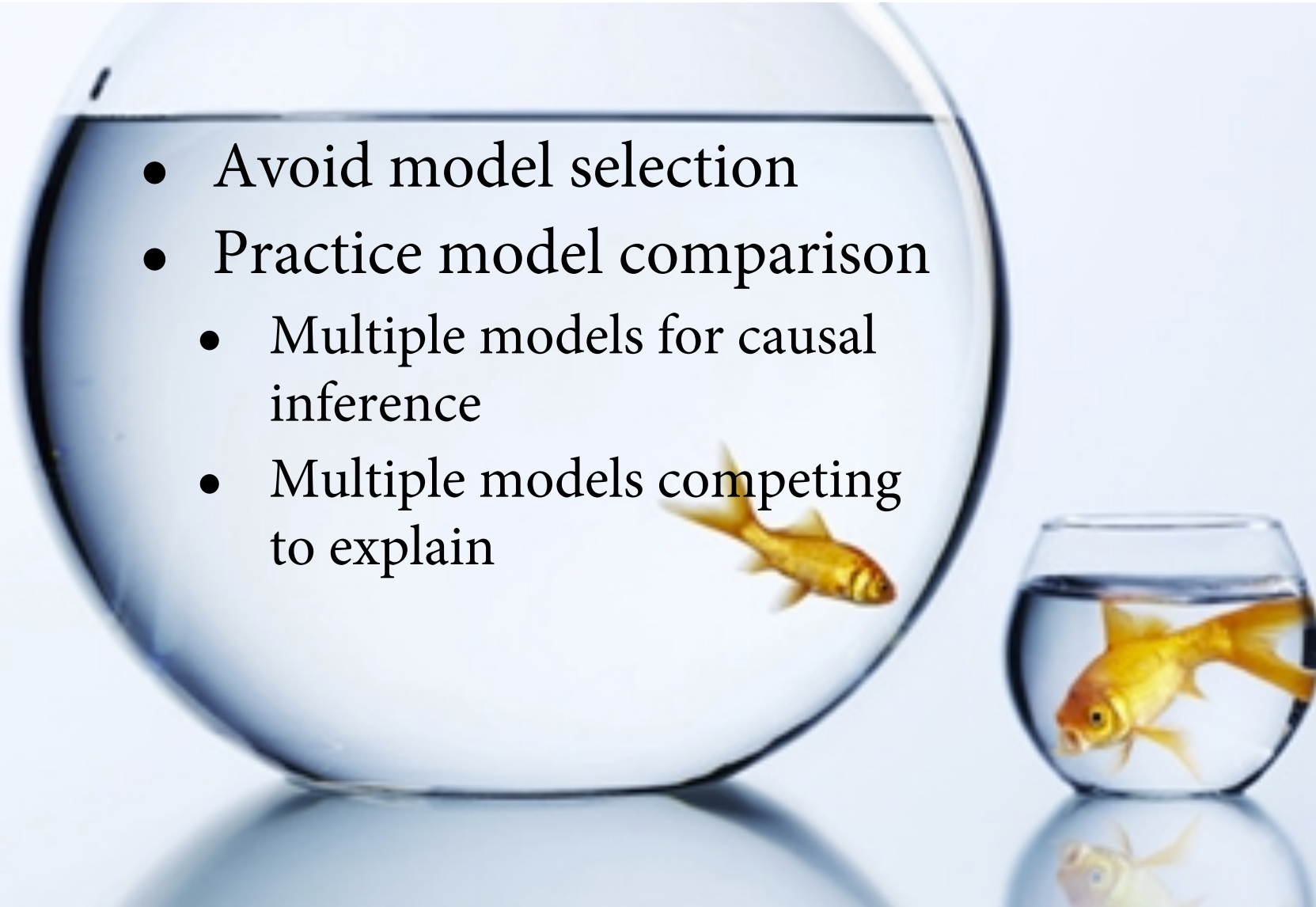
$$\text{WAIC}(y, \Theta) = -2(\text{lppd} - \underbrace{\sum_i \text{var}_{\Theta} \log p(y_i | \Theta)}_{\text{penalty term}})$$

- Does not assume Gaussian posterior
- WAIC function in rethinking



Using CV & WAIC

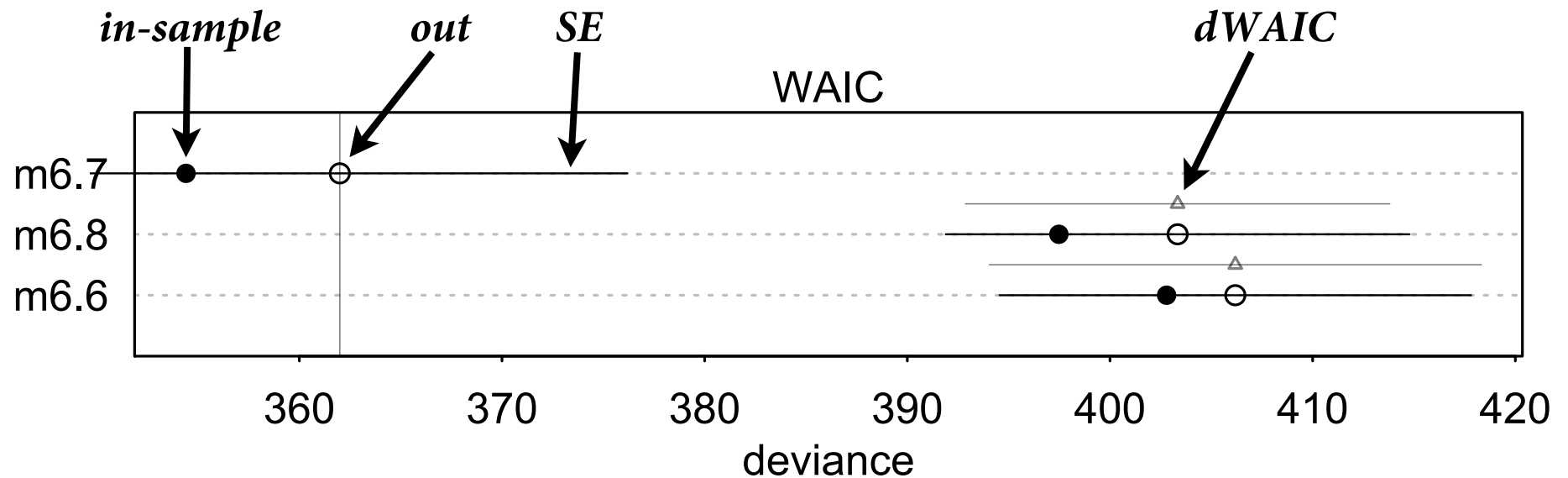
- Avoid model selection
- Practice model comparison
 - Multiple models for causal inference
 - Multiple models competing to explain



R code
7.27

```
set.seed(77)  
compare( m6.6 , m6.7 , m6.8 )
```

		WAIC	pWAIC	dWAIC	weight	SE	dSE
treat + fungus	m6.7	361.9	3.8	0.0	1	14.26	NA
fungus	m6.8	402.8	2.6	40.9	0	11.28	10.48
intercept	m6.6	405.9	1.6	44.0	0	11.66	12.23



↺↻ Maarten van Smeden retweetade



Maarten van Smeden

@MaartenvSmeden



We need Occam's taser: an automatic electric shock for anyone making their model unnecessarily complex

[Översätt tweeten](#)

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